

ECE 447

Lesson 11

Angle Modulation,

Part 1



UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
 - **Lab 3 Assignment.** Due Lesson 16 (Tue, 22 Sep) by 2359 via Gradescope upload.
 - HW2 coming next week
 - Skills Review graded. Working on the Labs and HW1...

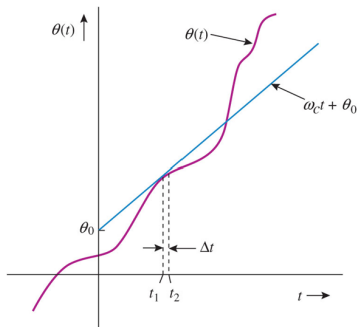
REVIEW

- DSB-SC: Problem 4.2-7 in textbook

ANGLE MODULATION

- Where is the message in FM or PM? How does that look mathematically?
- What is the relationship between frequency and phase?
Hint: in $\cos(\omega_c t + \theta_0)$, the phase $\theta_0 = \omega_c * \Delta t$ where Δt is the time shift
- Argument of a sinusoid is an angle (phase)
- If we plot $\omega_c t + \theta_0$ vs. t , we get a straight line
- $\frac{d(\omega_c t + \theta_0)}{dt} =$

ANGLE MODULATION



- Now consider a generalized sinusoidal signal: $\varphi(t) = A \cos \theta(t)$, where $\theta(t)$ is the generalized angle (not just a straight line)
- *Instantaneous frequency*: $\omega_i(t) = \frac{d\theta}{dt}$ and $\theta(t) = \int_{-\infty}^t \omega_i(\alpha) d\alpha$

ANGLE MODULATION

- PM

- $\theta(t) = \omega_c t + k_p m(t)$, where k_p is a constant
- $\varphi_{PM}(t) = A \cos[\omega_c t + k_p m(t)]$
- $\omega_i(t) = \frac{d\theta}{dt} = \omega_c + k_p \dot{m}(t)$

- FM

- ω_i varies linearly with $m(t)$ or $\omega_i(t) = \omega_c + k_f m(t)$, where k_f is a constant
- $\theta(t) = \int_{-\infty}^t \omega_i(\alpha) d\alpha =$
- $\varphi_{FM}(t) = A \cos[\omega_c t + k_f \int_{-\infty}^t m(\alpha) d\alpha]$

- Power of FM or PM signals?

COMPUTER SIMULATIONS

Jupyter Notebook files on Teams